Chapter 3 Notes

3.1

In the proof of Theorem 3.1, here is why the class of all transitive natural numbers is a set. We can show that it is a subset of ω, and is hence a set by A_2. For suppose x is a member of the class of all transitive natural numbers. Then obviously x is also a natural number, hence $x \in \omega$, thus completing the proof.

Note that here is why the only natural numbers allowed for by Definition 2.2 are the ones defined right before Definition 1.1. By Definition 2.2, a natural number is a member of every inductive set. Now, consider the following inductive set: $\emptyset$ is an element, and if x is an element, then so is x^+. Hence every natural number is a member of this set. But this set is just the set of all sets that were defined in the way just before Definition 1.1.

Exercise 3.1

Suppose that A is a class of natural numbers such that for every natural number n, if A contains all numbers less than n, then A contains n as well. Now, we can prove by induction that A contains all natural numbers. Consider the property P meaning A contains all numbers natural numbers less than or equal to n. Now, there are no natural numbers less than 0. Thus the statement that A contains all natural numbers less than 0 is vacuously true; hence A contains 0. Thus, because 0 is the only natural number less than or equal to 0, P holds for 0. Now suppose that P holds for some n, that is, A contains all natural numbers less than or equal to n. But then A contains all numbers less than $n+1$, meaning (by the definition of A) A also contains $n+1$. Thus, by induction, A contains all natural numbers.

Exercise 3.2

Consider a natural number n.

(a): Suppose that $x \in n$. Now, it cannot be that $n = \emptyset$. Thus n is non-empty, hence of the form $(n-1)^+$. Thus either $x \in n-1$ or $x = n-1$. Now, $\cup n$ is just the set of all elements of all elements of $n-1$ together with all elements of $n-1$. Now suppose $x = n-1$. Then $x \notin \cup n$. For suppose it were. Then it would either be an element of $n-1$ or it would be an element of an element of $n-1$. In the first case, we would have $n-1 \in n-1$, which is ruled out by Theorem 3.2. In the second case, there would be some y such that $x \in y$ and $y \in n-1$. But then, by the transitivity of $n-1$ (by Theorem 3.1), $x \in n-1$, that is, $n-1 \in n-1$, which is again ruled out by Theorem 3.2. Thus $n-1 \notin \cup n$, meaning the statement is not necessarily true.

It should be noted that all elements other than $n-1$ of n are members of $\cup n$. For all elements of n

are either elements of $n-1$ or equal to $n-1$. Thus, if they are not equal to $n-1$, the element is an element of $n-1$. But, because $n-1 \in n$, then all elements of $n-1$ are elements of $\cup n$.

(b): Suppose that $x \in \cup n$. Thus either $x \in n-1$ or x is an element of some element of $n-1$. In the first case, because $n-1 \in n$, $x \in n$ (by the transitivity of n). In the second case, because $n-1$ is transitive, it follows that $x \in n-1$. But then, because $n-1 \in n$ and n is transitive, $x \in n$. Thus, in either case, $x \in n$, and thus $\cup n \subseteq n$. Hence the statement is necessarily true.

(c): By part (b) of this problem, it follows that $x \in \cup n$ implies $x \in n$. But $n^+ = n \cup \{n\}$. Thus all elements of n are in n^+, meaning $x \in n^+$. On the other hand, suppose that $x \in n^+$. Then it could be that $x = n$. But then, if $x \in \cup n$, we would have either n being an element of an element of $n-1$, or $n \in n-1$. In the first case, the transitivity of $n-1$ would show that $n \in n-1$. Thus we would have, in either case, $n \in n-1$, which is impossible. For if $n \in n-1$, then, because $n-1 \in n$, $n-1 \in n-1$ by the transitivity of $n-1$, which is impossible (or, if $n-1 = \emptyset$, it could have no members). Hence it is not necessarily the case that $n^+ \subseteq \cup n$. Thus $\cup n = n^+$ does not necessarily hold.

Similar reasoning as to the above also shows that $n-1$ cannot be an element of $\cup n$. It should be noted that $n-1$ and n are the only members of n^+ that are not members of $\cup n$. For any other element of n^+ is an element of n but not equal to $n-1$. But $n = n-1 \cup \{n-1\}$. Hence it would be an element of $n-1$, and because $n-1$ is an element of n, it would also be an element of $\cup n$.

(d): On the one hand, suppose that $x \in \cup n^+$. Then either x is an element of an element of n, or x is an element of n. In the first case, because n is transitive, $x \in n$. The second case means $x \in n$. Thus, in any case, $x \in n$. On the other hand, suppose that $x \in n$. Then $x \in \cup n^+$, as $n^+ = n \cup \{n\}$. Hence $\cup n^+ = n$.

(e): By part (b) of this problem, $\cup n \subseteq n$ holds. But by the answer to part (a) of this problem, it does not necessarily follow that $n \subseteq \cup n$. Thus it is not necessarily true that $\cup n = n$.

Exercise 3.3

(a): On the one hand, suppose that $x \in \cup\omega$. Then x is an element of a natural number. Thus either $x \in n$ or $x = n$. Now we prove the following statement: every element of every natural number is a natural number. The base case vacuously holds, as $\emptyset$ has no elements. Now suppose the statement holds for some n. Then $n^+ = n \cup \{n\}$. Hence, if $y \in n^+$, either $y \in n$ or $y = n$. In the former case, by the induction assumption, it follows that y is a natural number, and in the latter case, y would be a natural number (as n is a natural number). Thus, in any case, y is a natural number, therefore completing the proof. As a consequence of having proven this statement, it follows that x is a natural number, and hence $x \in \omega$. On the other hand, suppose that $x \in \omega$. Thus, for some n, $x = n$. Now, all elements of n^+ are in $\cup\omega$. But n is an element of n^+. Thus $n \in \cup\omega$, and hence $x \in \cup\omega$. Thus we have proven that $\cup\omega = \omega$.

(b): On the one hand, suppose that $x \in \cup\omega^+$. Now, $\omega^+ = \omega \cup \{\omega\}$. Thus either x is an element of a natural number or x is a natural number. By the results of part (a) of this problem, it follows that if x is an element of a natural number, then x is a natural number. Thus x is a natural number, that is, $x \in \omega$. Now suppose that $x \in \omega$. Then $x \in \cup\omega^+$, as x would be an element of ω, which is an element of ω^+. Thus we have proven that $\cup\omega^+ = \omega$.

Note that a generalization cannot be made of this problem. Define $\{\omega_0, \omega_1, \omega_2, ..., \omega_n, ...\}$ in the following way: $\omega_0 = \omega$ and $\omega_{n+1} = \omega_n^+$. Then it does not follow that $\cup\omega_n = \omega$ for all n. For consider $n = 2$. This is ω_1^+, that is,

$$\omega_1 \cup \{\omega_1\} = (\omega \cup \{\omega\}) \cup \{(\omega \cup \{\omega\})\}$$

Thus $\omega \in \cup\omega_1^+$. But $\omega \notin \omega$, meaning $\cup\omega_2 \neq \omega$. Indeed, it is a general fact that, for $n \geq 2$, $\cup\omega_n \neq \omega$. To prove this, we can just prove that, for all $n \geq 2$, $\omega \in \omega_n$. We have just proven the base case. If the result

holds for some n, then $\omega \in \omega_{n+1}$, as $\omega_{n+1} = \omega_n \cup \{\omega_n\}$, and it was assumed that $\omega \in \omega_n$. Thus, for every $n \geq 2$, $\omega \in \omega_n$. Now we can prove the statement that, for $n \geq 2$, $\cup\omega_n \neq \omega$. We proved the base case for $n = 2$ above. Now consider some $n > 2$. Then $\omega_n = \omega_{n-1} \cup \{\omega_{n-1}\}$ and $\cup\omega_n$ consists of all elements of all elements of ω_{n-1} and all elements of ω_{n-1}. Because ω is an element of ω_{n-1}, it follows that $\omega \in \cup\omega_n$, and thus, because $\omega \notin \omega$, $\cup\omega_n \neq \omega$.

3.4

Consider an aritrary class A and an arbitrary function g. We shall say that A is inductive under g, or g-inductive, if the following two conditions hold:
(1) $\emptyset \in A$,
(2) for every x in A, the set $g(x)$ is in A.
We shall say that A is minimally inductive under g if A is inductive under g and if no proper subclass of A is inductive under g, in other words, any subclass of A that is inductive under g contains all elements of A.
Supppose M is minimally inductive under g and that R is a relation satisfying the following two conditions:
D_1 $R(x, 0)$ holds for every x in M;
D_2 for all x, y in M, if $R(x, y)$ and $R(y, x)$ then $R(x, g(y))$;
Then $R(x, y)$ holds for all x, y in M